

Enumeration of bacteria and cyanobacteria by flow cytometry

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Version 1.0

SUMMARY:

Seawater is collected and preserved to be analyzed in lab using an InFlux Flow Cytometer with the Spigot software package. The populations targeted for enumeration are *Prochlorococcus*, *Synechococcus*, unclassified pico-eukaryotes (cells sized around $1\mu\text{m}$), and heterotrophic non-fluorescing bacteria. Each sample is divided in two, with one aliquot analyzed for auto-fluorescence and the other for heterotrophic bacteria stained with SYBR Green I. In both cases, excitation is produced using stacked 488nm and 457nm lasers. The data is analyzed using FlowJo (v10) to determine cell concentrations (cells/mL).

1. Sample Collection

- 1.1 A 15mL polypropylene centrifuge tube is used to collect 10-12mL of seawater from the Niskin bottle
- 1.2 A 2.0mL subsample is taken and stored in a 2mL Corning cryovial (P/N 66021-974 from VWR) that has been pre-loaded with $30\mu\text{L}$ ($15\mu\text{L}$ per 1mL sample) of 16% paraformaldehyde (PFA) (P/N 15700 from Electron Microscopy Sciences).
- 1.3 The sample is mixed well (3x inversions of the cryovial) and stored in the dark at room temperature for 15 minutes.
- 1.4 Finally, the sample is flash frozen in liquid nitrogen and stored in a -80°C freezer.

2. Instrument Setup

- 2.1 Triple rinse and fill the sheath fluid reservoir with MilliQ Water
- 2.2 Attach a Millipore Sterivex 0.22um In-Line Filter (P/N: SVGPL10RC) to the InFlux intake for the sheath fluid.
- 2.3 The sheath pressure should be set to 20.0psi, and the sample pressure adjusted to yield a flow rate of approximately $30\mu\text{L}/\text{min}$ (normally 20.1-20.2psi). The C-MORE Hale InFlux uses a Sensirion SLI-0430 Flow Meter placed in-line between the sample line pinch valve and the nozzle assembly.
- 2.4 The InFlux interfaces with the software Spigot (ver. 6.1.10.0) from Cytopeia.

3. Laser Alignment

- 3.1 For HOT and SCOPE cell counts, the InFlux uses two lasers focused on the same point in the stream (both entering through the same pinhole). The laser sources are a 488nm 200mW laser (Coherent, 488-200 CRDH) and 457nm 300mW laser (CVI Melles Griot, 85-BLS-601), both operating at 100%.
- 3.2 Initial alignment is done using the 488nm laser, focusing the beam first on the Red and Orange channels and then on the FSC channel to maximize precision. The 457 laser alignment is done second, and is maximized to provide the highest signal on the Red and FSC channels.
- 3.3 Alignment is performed using Spherotech Ultra Rainbow Fluorescent Particles in the $1.0\text{-}1.5\mu\text{m}$ size (P/N: URFP-10-5). Generally, gains are adjusted such that the $1\mu\text{m}$ beads are tightly clustered near the $10^{3.1}$ decade for FSC and Red channels, the $10^{1.5}$ decade for the Orange channel, and the $10^{1.3}$ decade for the Green channel.
- 3.4 The optical pathway filters used for each channel are:
 - FSC: 488 Blocking (For cleaning the 488 light)
 - Red: 560LP > 610LP > 692/40BP
 - Orange: 560LP > (Reflect) 610LP > 585/40BP
 - Green: (Reflect) 560LP > 542/27BP

- 3.5 After final alignment, it is best to let the InFlux run for 45 minutes (while backflushing) to stabilize the stream. Moving forward, minor re-alignments may be necessary.

4. Sample Preparation

- 4.1 Samples are removed from -80°C storage after the InFlux has been fully aligned. The samples are placed on the bench at room temperature, in the dark to allow them to thaw.
- 4.2 Once thawed, the sample is briefly mixed by vortex (less than one second), and then split into two 1mL aliquots placed in 5mL polypropylene tubes (P/N: 352063 from Corning, henceforth PP tubes), one for unstained auto-fluorescence counts, and the other for DNA stained heterotroph counts. Once aliquoted, unstained are placed in a 4°C refrigerator in the dark.

4.3 Unstained Samples

- 4.3.1 2μL of a URFP-10-5 (1μm beads) dilution is added to every unstained sample, and briefly mixed by vortex.
- The bead dilution is prepared by adding 2 drops of the URFP-10-5 beads to 3mL of MilliQ water (in one of the PP tubes), and then briefly mixed by vortex.
 - The vortex used throughout this analysis is the Vortex Genie 2 from Scientific Industries, and the vortex speed is set to seven.
- 4.3.2 After the beads have been added, store the samples in a 4°C refrigerator in the dark.

4.4 Stained Samples

- 4.4.1 DNA Stained samples will not have the stain added until after the unstained samples have been run.
- 4.4.2 Preparation of the SYBR Green I working stock:
- 4.4.2.i Important Points:
- A working stock of SYBR Green I is made by a 50-fold dilution of the original SYBR concentrate (20μL SYBR Green I to 1000μL final volume, or a 20 SYBR : 980 Water ratio). The working stock is added to the samples at a dilution ratio of 200X (i.e. 5μL/mL sample).
 - When preparing the SYBR working stock, always use sterile or DNA-free consumables.
 - The DNA stain used in this analysis is Lonza SYBR Green I Nucleic Acid Gel Stain (P/N 50513). The SYBR Green I stock 10,000X concentrate is stored in a -20°C freezer with the included desiccant packet.
- 4.4.2.ii Remove the 10,000X SYBR Green I stock from the -20°C freezer, and place on ice to thaw.
- 4.4.2.iii Briefly (20 seconds) use a microcentrifuge to condense the SYBR Green I stock into before removing the aliquot. The centrifuge used here is a VWR Centrifuge Mini (P/N 10067-588).
- 4.4.2.iv Prepare the 200x working stock by diluting 20μL of SYBR Green I to 1000μL (20μL SYBR : 980μL MilliQ) using MilliQ water filtered through a 0.1μm Acrodisc syringe filter (VWR: P/N 28143-309). Mix briefly by vortex.
- 4.4.2.v SYBR Green I dilutions created in this manner will be stable at room temperature for one day, in a 4°C refrigerator for up to five days, and in a -20°C freezer for at least a month.
- 4.4.3 5μL of the SYBR Green I 200x working stock is added to each sample aliquot to be used for heterotroph counts, and briefly mixed by vortex.
- 4.4.4 Once the stain has been added to all aliquots, place them in a 4°C refrigerator for 20 minutes. Samples are stored in the 4°C refrigerator until analyzed.
- *Note:* SYBR plots (G vs FSC) often display a tighter and more intense Green signal if the stained samples are brought to room temperature (25 minutes on bench, in the dark), have the SYBR working stock added, and then are left to incubate (on the bench,

in the dark) for 20 minutes before being returned to the 4°C refrigerator. This practice does not alter the SYBR counts, but may ease downstream analysis.

5. Running Samples

- 5.1 Always check the instrument to make sure the data collected has low noise and few false positives in the areas of interest.
- 5.2 During the course of a normal run, the unstained sample aliquots will be run before the stained counterparts, because the SYBR Green leaves a residue in the sample line that must be cleaned out.
- 5.3 Throughout the run, it may be useful to occasionally take blank samples using MilliQ water (or, for stained samples, MilliQ water with 5 μ L SYBR Green working stock added). This will check for noise in the instrument, either electronic or due to deposits in the sample line.

5.4 Running Unstained Samples

- 5.4.1 Ensure that the InFlux is triggering on FSC with a trigger level in the 10-20 range.
- 5.4.2 Place the PP tube on the InFlux sampler, press Run, boost the sample until events appear on screen, and equilibrate the sample for 30+ seconds, until the event plots on Spigot are consistent.
- 5.4.3 Once the sample has equilibrated, the run. If you wish to cross compare the SLI-0430 Flow Meter with a mass difference method, remove the sample from the InFlux and get the initial mass.
- 5.4.4 Begin the list file collection in Spigot, reset the Totalizer on the SLI-0430 software, and press Run on the sample. Continue collecting data until the flow meter reads 100 μ L of sample have been run. Stop the run, remove the sample and weigh if desired.
- 5.4.5 Begin back flushing the InFlux sample line, this should be done for 30+ seconds between each sample.

5.5 Running Stained Samples

- 5.5.1 Change the trigger channel to Green, with a trigger level in the 40-50 range.
- 5.5.2 Prepare the stained samples for the run as described in Step 4.4, ensuring that the samples sit in the refrigerator for a minimum of 20 minutes before beginning the analysis.
- 5.5.3 Add 5 μ L of the SYBR Green I working stock to the remainder of an unstained sample, and run the sample to prepare the InFlux sample line for running stained samples.
- 5.5.4 Repeat the procedure for running unstained samples (Step 5.4), changing the analysis volume to 50 μ L from 100 μ L.

6. Machine Shutdown

- 6.1 During any run where SYBR Green I is used, the InFlux should first be cleaned by running 5% CONTRAD 70 detergent through the sample line, followed by 70% ethanol, followed by MilliQ water until the event rate for FSC detection is less than 20.

7. Data Analysis

- 7.1 The .fcs files are analyzed using FlowJo (ver. 10.2)
- 7.2 Gates are drawn for the following populations, with sequential gating indicated by the > symbol:
 - 7.2.1 *Prochlorococcus*: Red vs FSC > Orange vs FSC
 - 7.2.2 *Synechococcus*: Green vs Orange > Red vs FSC
 - 7.2.3 Pico-Eukaryotes: Red vs FSC > Orange vs FSC
 - 7.2.4 Heterotrophs* (SYBR, stained samples): Green vs FSC (*see below)
 - The final concentration for heterotroph concentrations must have the *Prochlorococcus* concentration subtracted from the SYBR concentration that was gated in the Green vs FSC plot. This is because both populations take up the DNA stain, and have similar FSC and SSC signals.

- 7.3 Once the raw counts have been extracted from FlowJo, the concentrations are determined by dividing the count by the corrected sample volume and converted to cells/mL.
 - 7.3.1 The measured volume (from the Flow Meter) must be corrected to account for dilution by PFA and beads or stain.
 - 7.3.2 Unstained volumes are divided by 1.017 (15 μ L PFA + 2 μ L beads) and stained volumes are divided by 1.020 (15 μ L PFA + 5 μ L stain)
 - 7.3.3 Conversion to seawater volume using the mass difference method is done using a static coefficient of 0.0009718, attained using the density of 1.026 μ g/ μ L for seawater.